CiteDrift: Do the Pages Cited by AI Search Answers Still Support Them?

Seungbum Ju
Samsung Electronics
Seoul, Republic of Korea
seungbum.ju@samsung.com

ABSTRACT

AI search engines back their statements with links, and readers keep those answers long after they were generated. A citation that supported a statement when it was made can later stop doing so, either because the link dies or because the page is rewritten. Link checkers see only the first case, and claim verifiers applied to a live citation see only the page as it is today. We demonstrate CiteDrift, a system that checks every statement–citation pair of an answer twice, against the page as it was when cited and as it is today, and reports whether support persisted, was lost with the link, or was lost through a change of content. CiteDrift runs entirely on one consumer GPU with open models. Re-checking 2,203 pairs from four generative search engines (2023) in October 2026, we find that 10.3% (95% CI 8.2–12.5%) of the checkable citations that annotators had judged fully supporting lost their link, a link-only checker reports 42% of these lost links as live, and CiteDrift flags a further 4.3% as content changes; an AI agent blind to our outcomes confirmed 22 of 37 (59%) of a sample of these flags. The demo offers an explorer of these results and a local mode that checks any pasted answer.

KEYWORDS

AI search, citation verification, reference rot, web evolution, claim verification

1 INTRODUCTION

Generative search engines answer a query with a few sentences and attach URLs to them. The links make an answer look checkable, and answers are kept and reused: they are shared as links, pasted into reports and stored in chat histories. Whether the cited page supports a statement, however, is a property of the page *at a given time*. The Web changes underneath citations: in scholarly articles, "one in five articles suffers from reference rot" [5] and "three out of four URI references lead to changed content" [4]. For AI search answers, verifiability has been measured once, at generation time, by human annotators [6] or by automatic auditors [10, 11].

Existing tools cover the two halves of the problem separately. Link checkers such as urlhealth [8] decide whether a cited URL resolves, but a page that still loads can have stopped saying what the answer claims. Claim verifiers and attribution evaluators [3, 9, 12] judge a statement against the document they are given; applied to a live citation, that is the page as it is today, so they cannot tell a citation that never supported a statement from one that did and then changed.

WWW Companion '27, May 10–14, 2027, Dublin, Ireland
2026. ACM ISBN 978-x-xxxx-xxxx-x/YY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

CiteDrift joins the two. For each statement–citation pair it judges support on the page as cited (an archived capture, or the page text saved with a dataset) and on the page today, and assigns one of six outcomes: *still supported*, *content changed*, *uncertain*, *link lost*, *could not check*, or *not supported when cited*. The contributions are: (1) a two-time, claim-level citation checker with a calibrated link-state step that separates bot walls and network filters from real loss and judges redirects by content; (2) a fully local implementation (no commercial API; one 12 GB GPU) with a web explorer and a check mode for any answer; and (3) a measurement: re-checking the human-annotated development and test citations of Liu et al. [6] after about three and a half years, 10.3% of the checkable citations judged fully supporting in 2023 lost their link (a link-only checker reports 42% of these as live), and CiteDrift flags another 4.3% as content changes. The demo relates to the Responsible Web track (information integrity, auditing), Search and Retrieval-Augmented AI (evaluation of AI search), and Web Mining (Web evolution).

2 SYSTEM

CiteDrift takes an answer with citation markers (`[1]`, `[2]`, …), the list of cited URLs and, optionally, the date of the answer. It splits the answer into sentences, pairs every sentence with the citations it carries, and processes each statement–citation pair in four steps (Fig. 1).

1. Link state. Each URL is requested once by an identified client with per-host spacing. A plain HTTP request is not enough to decide that a link is lost: in our measurement, 86 of the 193 URLs that answered a plain request with 401/403/429 and were re-opened opened normally in an ordinary browser, and 34 URLs were answered by our own network's web filter rather than by the site. CiteDrift therefore (i) recognises filter and bot-wall pages by their text, (ii) re-opens refused pages in an ordinary desktop browser (in the batch measurement for every refused page not answered by our network filter; on request in check mode); it does not solve challenges or bypass paywalls, and a page that appears only after a detected bot interstitial counts as a bot wall, and (iii) judges redirects by content, not by URL: a redirect counts as *moved* if the new page still contains at least 30% of the cited page's word 5-grams or keeps its title (thresholds set before the evaluation from pages that kept their URL and redirects to a site root; script in the repository), and as *redirected elsewhere* otherwise; without a "then" page only redirects to a site root count as lost. With this rule 126 of 186 redirects were moves. The resulting states fall into three groups: *reachable* (same URL, moved), *lost* (dead, error page, redirected elsewhere) and *unverifiable* (bot wall, network filter, server or TLS error, time-out, empty page, not requested).

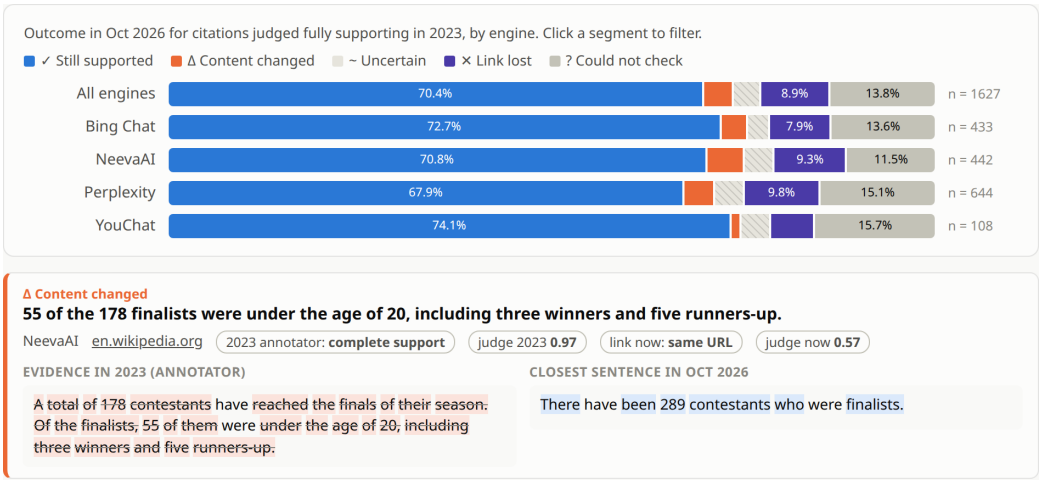


Figure 1: CiteDrift explorer. Top: outcome in October 2026 of all citations that annotators judged fully supporting in 2023, including those that could not be checked, per engine (segments filter the list). Bottom: one pair; the 2023 evidence and today's closest sentence are aligned word by word: the page is still online, some figures changed, and the judge's score dropped from 0.97 to 0.57.

2. Then and now text. The "now" page is reduced to its main content with Trafilatura [1]; the full visible DOM text is judged as well and the higher score counts, so that an extraction miss is not mistaken for a content change. The "then" page is the Internet Archive capture closest to the answer date, or page text saved with a dataset when available (for Liu et al., text extracted by the dataset authors' HTML-to-text pipeline). Archive lookups that fail (e.g., HTTP 429) are reported as such and never read as "the page did not exist".

3. Support judge. MiniCheck-Flan-T5-Large [9] (our re-implementation of its reference inference) scores the statement against every ~500-word chunk of a page and keeps the best chunk. Because AI search answers often copy phrases from their sources, the MiniCheck logit is combined with the share of the statement's word 4-grams found on the page (logistic regression fitted on the development split). The decision threshold is fixed on the development split and used unchanged at both time points.

4. Outcome. A pair is *not supported when cited* if the judge rejects the "then" page, *link lost* or *could not check* according to its link state, and *still supported* if the judge accepts today's page. Because the judge errs, a rejection today alone is not called a change: a pair is *content changed* only if, in addition, the passage that supported the statement then is no longer on today's page; if it is still there the pair stays still supported, and if the signals cannot be compared it is *uncertain*. For flagged pairs, an optional explainer (Qwen2.5-7B-Instruct [7], 4-bit) reads the best passage and writes a one-sentence reason.

Interface. A single web application (FastAPI back end, plain HTML/JS front end) has two modes. The *explorer* is a static page with our precomputed results on the Liu et al. data: per-engine outcome bars that act as filters, and one card per pair showing the statement, the annotator's 2023 evidence next to the closest sentence on today's page with a word-level diff, link state and both judge scores. The *check* mode runs the full pipeline on a pasted answer on the user's own machine and streams each outcome as soon as its page has been fetched and judged; nothing the user submits leaves the machine except requests for the cited pages, their robots.txt files and their archive records.

3 EVALUATION

Data. Liu et al. [6] collected answers of Bing Chat, NeevaAI, Perplexity and YouChat between late February and late March 2023 and had annotators judge each statement–citation pair (complete, partial or no support), copying evidence sentences from the page. The release also contains the text of each cited page, released with the dataset (extracted with DOM Distiller and Trafilatura); we use it as the "then" page. We develop and report on the development (1,106 pairs) and test (1,097) splits, 2,203 pairs and 1,473 distinct URLs re-fetched on 15 October 2026, and keep the training split (8,834 pairs) as an untouched held-out check, re-fetched about 14 hours later. After calibration, 1,105 URLs (75.0%) were reachable, 142 (9.6%) lost and 226 (15.3%) could not be checked from our network.

E1: judge accuracy. Table 1 compares judges on the 2023 text against the annotators' labels (target: complete support; threshold chosen on dev, reported on test). MiniCheck alone and word 4-gram overlap are close on average because these engines often copy phrases from their sources, but on the 57% of test statements that share few 4-grams with the page, MiniCheck (AUC 0.78) is far better than string matching (0.65). Their combination is best overall. Qwen2.5-7B (4-bit) on the three best chunks accepts 85% of all pairs and is weaker as a verifier; we use it only to explain flags.

E2: drift 2023→2026. We take the 1,627 pairs that annotators judged *fully* supporting; 224 of them could not be checked, leaving 1,403. With the annotators' evidence available, a reachable pair that the judge rejects today counts as *content changed* only if the judge accepted the 2023 text *and* not all of the annotator's evidence is still on today's page; if all of it is still there it stays still supported (115 pairs), otherwise it is *uncertain*. 144 citations (10.3%; 95% CI

Table 1: E1: support judges on the 2023 page text vs. the annotators' labels (test split, n =1,097; 95% bootstrap CIs).

Judge	AUC	Bal. acc.	$F1_{unsup.}$
Word 4-gram overlap	0.78	0.73	0.59
MiniCheck-Flan-T5-L	0.82	0.75	0.63
Qwen2.5-7B (4-bit), top-3 chunks	–	0.64	0.45
MiniCheck + overlap (ours)	**0.86**	**0.78**	**0.67**
95% CI	0.83–0.88	0.75–0.81	

Table 2: E2: outcome in Oct 2026 of citations judged fully supporting in 2023 (share of checkable pairs), and how well each method finds lost support (estimated over all 1,403 checkable pairs: spot-check sample weighted by stratum, the 82 pairs on dead or error pages counted as lost).

Engine	checkable	link lost	content changed
Bing Chat	374	9.1%	3.7%
NeevaAI	391	10.5%	5.4%
Perplexity	547	11.5%	4.4%
YouChat	91	6.6%	1.1%
Method	precision	recall	F1
urlhealth (not LIVE)	0.54	0.24	0.33
String match (4-gram)	0.43	0.77	0.55
CiteDrift	**0.85**	0.44	0.58

8.2–12.5%, bootstrap over URLs) lost the link, 60 (4.3%, CI 3.1–5.5%) are flagged as content changes, and 54 (3.8%) are uncertain (Table 2). On the test split alone, which played no part in fitting the judge (the outcome rule was chosen on a spot-check sample that included test pairs), the rates are 9.9% and 4.0%. On the held-out training split, processed the same way after all design choices were fixed, the judge reaches AUC 0.86 on the 2023 text, 10.0% of 5,611 checkable citations lost the link and 3.4% are flagged as content changes; a fresh blind AI agent confirmed 29 of 44 (66%) of the decidable flags in a sample of 50.

Validation. We have no human labels for 2026, so AI agents (Claude, a fresh context for each sample), blind to all methods and shown each statement, the annotator's evidence, both URLs and the page text, labelled stratified random samples. On the first (100 items), only 21 of 43 (49%) content changes flagged by a looser rule (either signal alone) were confirmed, so we adopted the two-signal rule; we report the second, fresh sample (128 items). There the agent confirmed 22 of 37 (59%) flagged content changes as no longer fully supported, 15 of 17 (88%) "redirected elsewhere" links as leading to a different document, and 24 of 29 (83%) "still supported" pairs. Because the agents and the 2023 annotators can simply disagree, another agent labelled the same pairs on the *2023* text: it already judged 7% of the still-supported pairs and 8% of the flagged changes not fully supported in 2023 (vs. 17% and 58% today; 29 and 36 pairs labelled both times). The flagged changes are thus mostly real changes rather than labeller disagreement. Netting out the disagreement and weighting by stratum size, content changes remove full support from about 12% (CI 3–24%) of the reachable citations, possibly more than CiteDrift flags; it rests mostly on the few sampled still-supported pairs judged changed. The two-signal rule trades recall for precision. The author re-checked 20 of the second sample's items by hand and agreed with 58 of 60 (97%) of the agent's labels.

Baselines. urlhealth [8] reports 60 of the 144 lost links as live, nearly all of them redirects to a home page or another document, and calls 7 still-supported citations "likely hallucinated". It labelled 54 URLs "likely hallucinated" although annotators read all of them in 2023; 8 of them load normally for CiteDrift, 8 were answered by our network filter, a later re-query found archive records for 41, and its code turns any failed archive lookup into "no snapshot". Over all checkable citations (Table 2), urlhealth misses most lost support. String matching on today's page (same text views as CiteDrift) finds the most, but more than half of its flags are false (precision 0.43), likely because answers paraphrase; CiteDrift is the conservative option (precision 0.85, CI 0.78–0.90), at the cost of recall; the two F1 scores do not differ reliably, and the reference (the agent's 2026 label) includes labeller disagreement, which lowers every method's recall.

E3–E4: cost and check mode. On one RTX 5070 (12 GB), judging one statement against both views of today's page takes a median 291 ms. Checking a whole answer (20 test answers; median 2 URLs, archive capture as "then") takes a median 92 s, of which 97% is spent waiting on polite page and archive requests and 1.5 s on judging; peak GPU memory is 3.4 GB (9.1 GB with the explainer, measured on five answers). For 37 of the 73 statement–citation pairs no archived "then" page could be retrieved (35 lookup errors, mostly without a logged cause, and 2 refused capture fetches); such pairs are marked as having no "then" page, and on reachable pages they are labelled still supported if today's page supports them and uncertain otherwise. Run on 60 test answers with the dataset's 2023 text, check mode, run without its optional browser re-check, agreed with the E2 tool-mode outcome on 93% of the 166 pairs it could match to E2 (of 225; 89% without pairs both judged unsupported in 2023); most differences are pages that the batch run had opened in a browser.

Limitations. The "then" text was extracted differently from today's; outcomes depend on one network vantage point and fetch time; 224 supported citations could not be checked; and the 2026 validation rests on an AI labeller. Outcomes are therefore shown with scores and evidence for the reader to inspect.

4 DEMONSTRATION

At the venue we bring a machine with one consumer GPU (all measurements here are on a 12 GB desktop RTX 5070) running the full system: the explorer works offline, check mode needs the venue network. Visitors can (1) browse the 2023→2026 explorer, click an engine's "content changed" segment and read, side by side, the sentence an annotator copied in 2023 and what the page says today, or list the 117 citations that lost support although urlhealth reports their link as live; (2) paste an answer they obtained from any AI search engine during the conference, optionally with a past date, and see each citation's outcome appear, with both judge scores and the passages that decided it, as its page is fetched; and (3) toggle the explainer to see why a citation was flagged. Visitors without a GPU can replay recorded checks of five 2023 answers in the public

explorer. The code is at https://github.com/seungbum/citedrift, the public explorer at https://seungbum.github.io/citedrift/, and a two-minute video walk-through at https://seungbum.github.io/citedrift/citedrift_demo.mp4.

5 RELATED WORK

Reference rot. Studies of scholarly articles separate link rot from content drift and find both common [4, 5]; CiteDrift measures both for AI search citations, per statement rather than per URL. **Citation and attribution checking.** Liu et al. [6] audited four generative search engines with annotators; SourceCheckup [11] and DeepTRACE [10] automate such audits with LLM judges; ALCE [3] and AttrScore [12] evaluate attribution automatically; Facts&Evidence [2] checks claims against multiple evidence sources. All judge support at one point in time. **Link checking and freshness.** urlhealth [8] classifies cited URLs as live, stale or likely hallucinated using the Wayback Machine, without looking at content. OwlerLite [13] detects page changes to keep a retrieval index fresh before answering; CiteDrift audits citations after an answer was given.

6 ETHICAL USE OF DATA

We use only public data, the verifiability judgments of Liu et al. [6] (MIT licence); no human participants were involved, and the 2026 spot checks were labelled by AI agents on locally stored text. Requests to cited sites: (i) our identified client, once per URL for the measurement (at least five seconds between requests to a host), and once per run for the pages of the 20 E3 answers (three full runs; explainer runs on one and twice on five answers), the 60 E4 answers (four runs) and example answers citing six sites (about twenty test and video runs, plus a few pilot and connectivity tests), two seconds apart per host (a host's robots.txt and first page request were unspaced until a late fix); sites that ask automated clients not to fetch a page were never requested (redirect targets were not re-checked); (ii) one visit with an ordinary browser under automation for 1,463 refused, failing or empty pages (94 twice; 118 dead training-split pages opened under a broader rule are not used), without solving challenges, logging in or bypassing paywalls; pages reached only after a detected bot interstitial count as bot walls; (iii) one urlhealth check per fetchable development/test URL, run as published (HEAD, a GET after some errors, an archive lookup after a 404; browser user agent), rate-limited like all our archive lookups. Page bodies stay local: the public explorer shows the dataset's evidence and at most one current-page sentence (up to 300 characters) per pair. Check mode processes the answer locally, stores nothing, and requests only the cited pages, their robots.txt files and archive records (and, on request, one browser visit; one such test was run).

ACKNOWLEDGMENTS

Generative AI disclosure. We used a generative AI assistant (Anthropic Claude) to implement the system and the experiment scripts, run the experiments, produce the 2026 validation labels (as stated in Section 3), collect bibliographic metadata, draft the text, and script the video voice-over, which is synthetic speech from the open Kokoro-82M model. The author defined the problem formulation and venue scope, directed key design decisions (including the two-signal drift threshold), independently conducted the author hand-check on the validation sample (Section 3), interpreted the empirical findings, and reviewed and revised the manuscript, taking full responsibility for all content and conclusions.

REFERENCES

[1] Adrien Barbaresi. 2021. Trafilatura: A Web Scraping Library and Command-Line Tool for Text Discovery and Extraction. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing: System Demonstrations*. Association for Computational Linguistics, 122–131. https://doi.org/10.18653/v1/2021.acl-demo.15

[2] Varich Boonsanong, Vidhisha Balachandran, Xiaochuang Han, Shangbin Feng, Lucy Lu Wang, and Yulia Tsvetkov. 2025. FACTS&EVIDENCE: An Interactive Tool for Transparent Fine-Grained Factual Verification of Machine-Generated Text. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (System Demonstrations)*. Association for Computational Linguistics, 437–448. https://doi.org/10.18653/v1/2025.naacl-demo.35

[3] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling Large Language Models to Generate Text with Citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 6465–6488. https://doi.org/10.18653/v1/2023.emnlp-main.398

[4] Shawn M. Jones, Herbert Van de Sompel, Harihar Shankar, Martin Klein, Richard Tobin, and Claire Grover. 2016. Scholarly Context Adrift: Three out of Four URI References Lead to Changed Content. *PLOS ONE* 11, 12 (Dec. 2016), e0167475. https://doi.org/10.1371/journal.pone.0167475

[5] Martin Klein, Herbert Van de Sompel, Robert Sanderson, Harihar Shankar, Lyudmila Balakireva, Ke Zhou, and Richard Tobin. 2014. Scholarly Context Not Found: One in Five Articles Suffers from Reference Rot. *PLoS ONE* 9, 12 (Dec. 2014), e115253. https://doi.org/10.1371/journal.pone.0115253

[6] Nelson Liu, Tianyi Zhang, and Percy Liang. 2023. Evaluating Verifiability in Generative Search Engines. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. Association for Computational Linguistics, 7001–7025. https://doi.org/10.18653/v1/2023.findings-emnlp.467

[7] Qwen Team. 2025. Qwen2.5 Technical Report. arXiv:2412.15115 [cs.CL] https://arxiv.org/abs/2412.15115

[8] Delip Rao, Eric Wong, and Chris Callison-Burch. 2026. Detecting and Correcting Reference Hallucinations in Commercial LLMs and Deep Research Agents. arXiv:2604.03173 [cs.CL] https://arxiv.org/abs/2604.03173

[9] Liyan Tang, Philippe Laban, and Greg Durrett. 2024. MiniCheck: Efficient Fact-Checking of LLMs on Grounding Documents. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 8818–8847. https://doi.org/10.18653/v1/2024.emnlp-main.499

[10] Pranav Narayanan Venkit, Philippe Laban, Yilun Zhou, Kung-Hsiang Huang, Yixin Mao, and Chien-Sheng Wu. 2025. DeepTRACE: Auditing Deep Research AI Systems for Tracking Reliability Across Citations and Evidence. arXiv:2509.04499 [cs.CL] https://arxiv.org/abs/2509.04499

[11] Kevin Wu, Eric Wu, Kevin Wei, Angela Zhang, Allison Casasola, Teresa Nguyen, Sith Riantawan, Patricia Shi, Daniel Ho, and James Zou. 2025. An automated framework for assessing how well LLMs cite relevant medical references. *Nature Communications* 16, 1 (April 2025). https://doi.org/10.1038/s41467-025-58551-6

[12] Xiang Yue, Boshi Wang, Ziru Chen, Kai Zhang, Yu Su, and Huan Sun. 2023. Automatic Evaluation of Attribution by Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. Association for Computational Linguistics, 4615–4635. https://doi.org/10.18653/v1/2023.findings-emnlp.307

[13] Saber Zerhoudi, Michael Dinzinger, Michael Granitzer, and Jelena Mitrovic. 2026. OwlerLite: Scope- and Freshness-Aware Web Retrieval for LLM Assistants. In *Companion Proceedings of the ACM Web Conference 2026*. ACM, 216–220. https://doi.org/10.1145/3774905.3793140